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706. Design Hashmap

↳ Fn → void put (int key, int value)
→ int get (int key)
void remove (key)

Brute Force key = 10^6.

Take array of size 10^6+1.

Ex. put (2,2)

0	1	2	. 10^6
		2	

- T.C = O(1)
S.C = 10^6

Probⁿ - If key = int / string → Probⁿ (v)

Optimal Approach

In Hash map:

key —(hash Fn)→ index
↳ (put value on that index).

Simple Hash Fn - take modulo (%)

ex. (10, 2) → 10 % 2 = 0
(20, 2) → 20 % 2 = 0
] - collⁿ probⁿ
↓ solved by
1) chaining (v)
2) open addressing.

Chaining

↳ probⁿ - all key should not come in one chain. For this

Load Factor:

Solⁿ: Take Arraylist of Arraylist (int, int).

vector < list < pair < int, int > > > bucket (10^4);

put (key, value) {

bucket_no = key % size;

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```
chain = bucket [bucket_no.]

    for (auto & it ; chain) {
        if (it.first == key) {
            it.second = value
            return;
        }
    }
    chain.emplace_back (key, value);

}
get (key) {
    bucket_no = key % size;
    chain = bucket [bucket_no];
    apply for loop & search  — if found — return value
                             — else return -1

}
remove (key) {
    1) find bucket_no.
    2) find chain
    3) iterate on chain  — if key found - chain.erase(it)
                         — it++
```